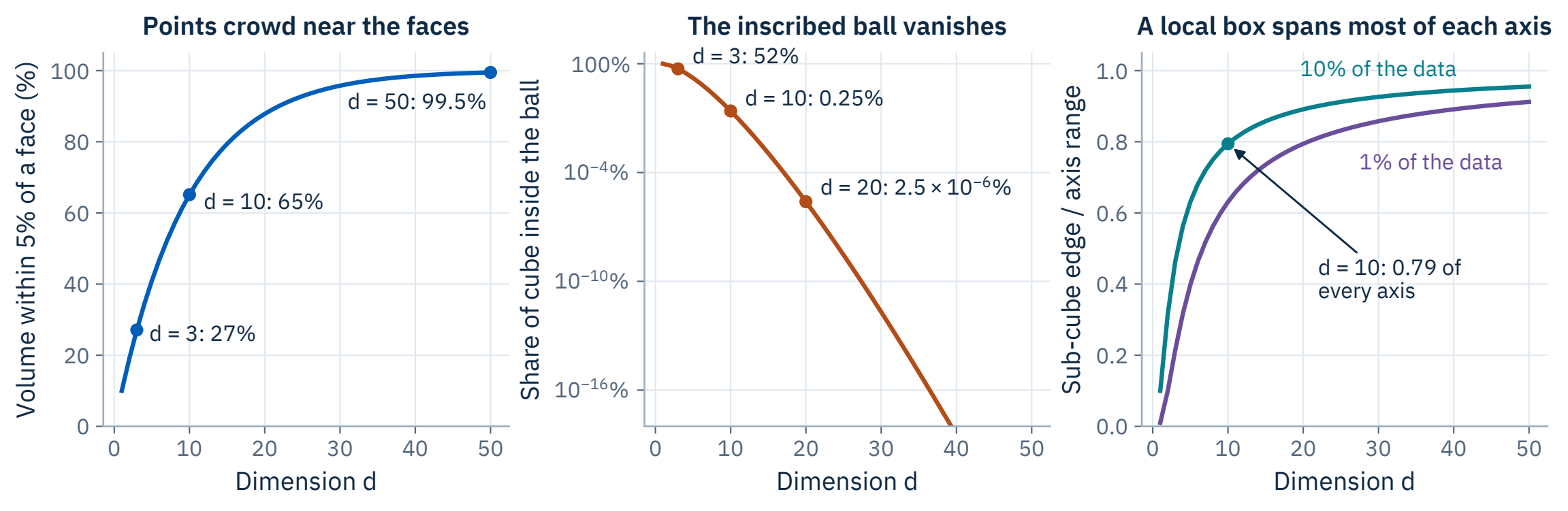
Points crowd near the faces
Volume within 5% of a face (%)
100
80
60
40
20
0
d = 50: 99.5%
d = 10: 65%
d = 3: 27%
0
10
20
30
40
50
Dimension d
The inscribed ball vanishes
Share of cube inside the ball
100%
10^{-4}%
10^{-10}%
10^{-16}%
d = 3: 52%
d = 10: 0.25%
d = 20: 2.5×10^{-6}%
0
10
20
30
40
50
Dimension d
A local box spans most of each axis
Sub-cube edge / axis range
1.0
0.8
0.6
0.4
0.2
0.0
10% of the data
1% of the data
d = 10: 0.79 of
every axis
0
10
20
30
40
50
Dimension d